

LYCEUM OF THE PHILIPPINES UNIVERSITY - CAVITE

**ATLAS: ADVANCED THESIS LOGISTICS AND AI SYSTEM FOR
COECSA AT LYCEUM OF THE PHILIPPINES UNIVERSITY CAVITE**

An Undergraduate Thesis
Submitted to the Faculty of the
College of Engineering, Computer Studies and Architecture
Lyceum of the Philippines University-Cavite

In Partial Fulfillment
of the Requirements for the Degree of
Bachelor of Science in Computer Science
with specialization in Web and Mobile Technology

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COLLEGE OF ALLIED MEDICAL SCIENCES

LYCEUM OF THE PHILIPPINES UNIVERSITY - CAVITE

ACCESS LEAF

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with Specialization in Web and Mobile Technology**

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**ATLAS: ADVANCED THESIS LOGISITICS AND AI SYSTEM FOR
COECSA OF LYCEUM OF THE PHILIPPINES UNIVERSITY CAVITE**

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We are extremely grateful to all those who have supported and contributed to the completion of this thesis that marks a significant milestone in my academic journey.

Allow me to thank a few people who made this success possible.

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LIST OF ABBREVIATIONS

DNA	Deoxyribonucleic acid
PCR	Polymerase chain reaction
RNA	Ribonucleic acid

ABSTRACT

Thesis defense management and evaluation processes within the academic institution are often challenged by scheduling conflicts, inconsistent assessments and limited tools to ensure originality in research titles. This study aims to develop ATLAS (Advanced Thesis Logistics and AI System), a web-based system that utilizes AI to streamline scheduling, improve assessment accuracy, and promote academic integrity within the College of Engineering, Computer Studies, and Architecture (COECSA) at Lyceum of the Philippines University Cavite (LPU-C). ATLAS integrates the Genetic Algorithm for an optimized scheduling system and Gemini, a Large Language Model (LLM), to aid in plagiarism detection and sentiment analysis for thesis feedback. Data from student researchers, thesis advisers, and program chairs within COECSA were collected to evaluate system effectiveness based on ISO/IEC 25010 quality standards, covering functionality, usability, and reliability. **TBC**

Keywords: thesis defense management, web-based system

CHAPTER I
INTRODUCTION

Background and Rationale of the Study

The Lyceum of the Philippines University-Cavite (LPU-C), the College of Engineering, Computer Studies, and Architecture (COECSA), faces logistical challenges in managing research presentations for its growing student population. The current manual system for scheduling defenses, evaluating theses, and organizing documents results in inefficiencies, time conflicts, and delayed feedback. The increasing demand for streamlined academic processes highlights the need for an AI-driven system to optimize scheduling, automate assessments, and enhance overall transparency. The proposed ATLAS system addresses these issues by utilizing modern technologies to improve both faculty and student experiences in managing research presentations. Currently, (Leaño et al., 2013) employs a manual system for the managing capstone and research presentations. This traditional process involves time-consuming tasks such as scheduling, which requires coordinating the availability of students, advisers, and panelists, and often results in conflicts or delays. The grading system, similarly, relies on manual input from the panelists, making it prone to errors and inefficiencies. The management of research projects, from recording schedules to tracking grades, also lacks a centralized, automated platform, further complicating the process as the student

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	population grows. This manual approach leads to increased administrative workload and delays in project assessments. Supporting the need for an AI-driven system, research highlights several key challenges in academic processes. A study by (Hariyono et al., 2021) found that students often struggle to select relevant thesis topics aligned with the current research trends, leading to less impactful research. By identifying topic themes such as neural networks and software engineering practices, the study emphasized the importance of guiding students towards appropriate topics. Additionally, (Nasien et al., 2022) demonstrated the effectiveness of genetic algorithms in optimizing course scheduling, reducing conflicts and improving resource allocation. These studies underscore the benefits of automation in academic settings, ensuring efficiency and enhancing student outcomes. The study aims to address several challenges faced by students and faculty in managing thesis projects and research presentations. By developing the ATLAS: Advanced Thesis Logistics and AI System, the study seeks to solve issues related to inefficient scheduling, difficulty in selecting relevant research topics, and inconsistent grading processes. The system will streamline scheduling through an algorithm, ensure the uniqueness of research titles, and track document completion. Additionally, ATLAS will provide AI-driven feedback using the Gemini Pro LLM, improving both the efficiency and accuracy of thesis evaluations while reducing human error.	
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Objectives of the Study

The study aims to develop ATLAS: Advanced Thesis Logistics and AI System for LPU-Cavite COECSA Department.

Specifically, the project will:

1. Develop ATLAS: Advance Thesis Logistics and AI System for LPU-Cavite COECSA Department with the use of the following features.

a) Thesis Topic Decision Tool

- This tool utilizes an API or an internal database to suggest relevant research topics based on current academic trends, ensuring quality thesis projects.

b) Research Title Acceptance Tool

- This tool assesses the uniqueness of a title using sources from open-access repositories and analyses them with the Gemini Pro LLM Model.

c) Scheduling System

- Aligns the defense schedule of both students and panelists based on availability. A feature which uses the Genetic Algorithm.

d) Content Management System

- Allows modular modification of rubrics, team lists, title lists, and the panelists assignments and the overall looks of the system.

- e) Requirement Checker Tool
 - Helps students and advisers track the completion of necessary documents before the defense.
- f) Pass Recommendation System
 - Utilizes the Gemini Pro LLM to process scores, comments, and the paper’s sentiment of the panelists to generate automated, suggestive feedback, improving consistency and reducing errors.

- 2. The project will use HTML/CSS and JavaScript for front-end flexibility and a user-friendly interface. PHP will be used for backend to manage server-side logic and real-time data processing. For database management, MySQL will be utilized to provide scalable and efficient data storage.
- 3. Evaluate and improve the system using ISO/IECC 25010 and Mobile Application Rating Scale (MARS).

Significance of the Study

The development of ATLAS: Advanced Thesis Logistics and AI System for LPU-Cavite COECSA Department aims to change how research presentations are scheduled, evaluated and managed. This study is important as it introduces an AI-driven system that rationalizes the logistical challenges faced by both students and

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	school faculty in organizing thesis defenses, while improving the overall efficiency and transparency of the academic process. Student Researchers of LPU-C will greatly benefit from this system. ATLAS offers students clear and organized scheduling, performance assessments, and presentation requirements. Our system improves the research and defense processes accountability and transparency. Thesis and Capstone Advisers will also experience significant advantages. ATLAS will provide them an efficient platform for scheduling optimizing time management, research presentations, and minimizing conflicts. Using our system, advisers may monitor the students' area of improvement and make well informed decisions. Research Teachers of LPU-C COECSA will benefit from the system's AI capabilities in facilitating research title validation, plagiarism detection, and communication between students and faculty. By implementing our system LPU-C's commitment on using modern technologies to improve academic processes. Future Researchers stand to benefit from this study as it can serve as a model for incorporating AI into academic scheduling and assessment systems. Future Researchers could utilize this framework to improve and innovate the academic scheduling and evaluation system.	
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	Scope and Limitation The scope of this study is focused on developing and implementing ATLAS: Advanced Thesis Logistics and AI System specifically for the College of Engineering, Computer Studies and Architecture (COESCA) at Lyceum of the Philippines University-Cavite (LPU-C). The system aims to enhance the processes of scheduling, grading, and managing capstone and research presentations. Key features of the system include the Thesis Topic Decision Tool, Research Title Acceptance Tool (utilizing Gemini Pro LLM), Scheduling System, Content Management System, Requirement Checker Tool, and Pass Recommendation System. The study will be conducted at LPU-C and will focus on addressing the logistical and academic challenges in the COESCA, with data collection taking place over the course of one semester. The system will only be applicable within the COESCA, and its features are tailored to the needs of the department, which may not be transferable to other departments without modification. The study will involve key stakeholders including students, research advisers, and panelists. It will not cover broader administrative activities, plagiarism checks beyond title comparison, or other academic programs within the university. Additionally, while AI-driven tools like the Gemini Pro LLM will assist in the decision-making process, human intervention remains essential for final grading and evaluation to ensure fairness and accuracy. The system will require an internet connection, compatible devices and is designed for web-based access.	
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	effectiveness in solving complex optimization problems across various fields. Their application ranges from machine learning to biomedical research, showcasing their adaptability and robustness in diverse scenarios. One prominent application of Genetic Algorithm is in the optimization of machine learning models. For instance, Wang (2023) developed an automatic grading system for English compositions that integrates a Genetic Algorithm with a Convolutional Neural Network (CNN). This approach enhances grading accuracy and optimizes the evaluation process by continuously refining the model through genetic operations such as selection, crossover, and mutation (Wang ,2023). Genetics Algorithm are used within ATLAS for optimization tasks, particularly in enhancing scheduling and resource management. Inspired by natural selection, genetic algorithms iteratively search for the best solutions to complex problems. In ATLAS, they can be applied to optimize schedules, ensuring that defense presentations are arranged with minimal conflicts by efficiently allocating time slots based on the availability of students, advisers, and panelists. This not only reduces scheduling conflicts but also improves the overall efficiency of the system, making it more adaptable to the growing number of students and faculty involved in research.	
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	importance of optimizing their architectures to enhance their practical usability in diverse applications (Cai, 2023). The study highlights the ethical and academic challenges of students using Large Language Models (LLMs) to write theses, including concerns over academic integrity, lack of authentic learning, and the risk of passing off AI-generated work as their own. While guidelines were proposed to address these issues, our thesis improves on this by integrating AI responsibly. Through tools like the Thesis Topic Decision Tool and Research Title Acceptance Tool, our system supports students in selecting relevant and unique research topics while ensuring originality. Additionally, the Pass Recommendation System ensures that AI aids decision-making without replacing critical human evaluation, safeguarding both academic integrity and learning outcomes. (Leary ,2023) LLM such as the Gemini Pro LLM used in ATLAS, adds a layer of AI-driven intelligence for thesis evaluations and feedback. LLMs can analyze text data, such as research titles or student-submitted documents, to provide relevant feedback, detect potential issues, and suggest improvements. For instance, the LLM helps in assessing the uniqueness of thesis titles by comparing them with existing works and generating recommendations to avoid redundancy or plagiarism. The use of LLM in ATLAS ensures that evaluations are more consistent and precise, as it supports the decision-making process with AI-driven insights, while human reviewers still maintain control over the final evaluations.	
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	scheduling in universities, (Chen et al., 2021) focused on optimizing class schedules through automated conflict resolution and efficient resource management. Their work highlights the importance of minimizing scheduling conflicts to enhance the educational experience for both students and faculty. While their study employed traditional software algorithms to achieve these goals, our system applies AI-driven scheduling algorithms to further enhance the accuracy and efficiency of the scheduling process. This approach not only streamlines the scheduling of research presentations but also optimizes panelist and student availability, offering a more flexible and responsive solution to scheduling challenges. Automated Scheduling System is another critical component that simplifies the logistics of research presentations and defenses. By leveraging AI and algorithms, this system automatically aligns the schedules of students, advisers, and panelists, ensuring that conflicts are minimized and that presentations can proceed smoothly. It takes into account various constraints, such as availability, deadlines, and institutional requirements, to generate the most efficient and conflict-free schedule possible. This automated process reduces the manual workload on administrators and minimizes errors, making it easier to manage multiple research defenses in a growing academic environment.	
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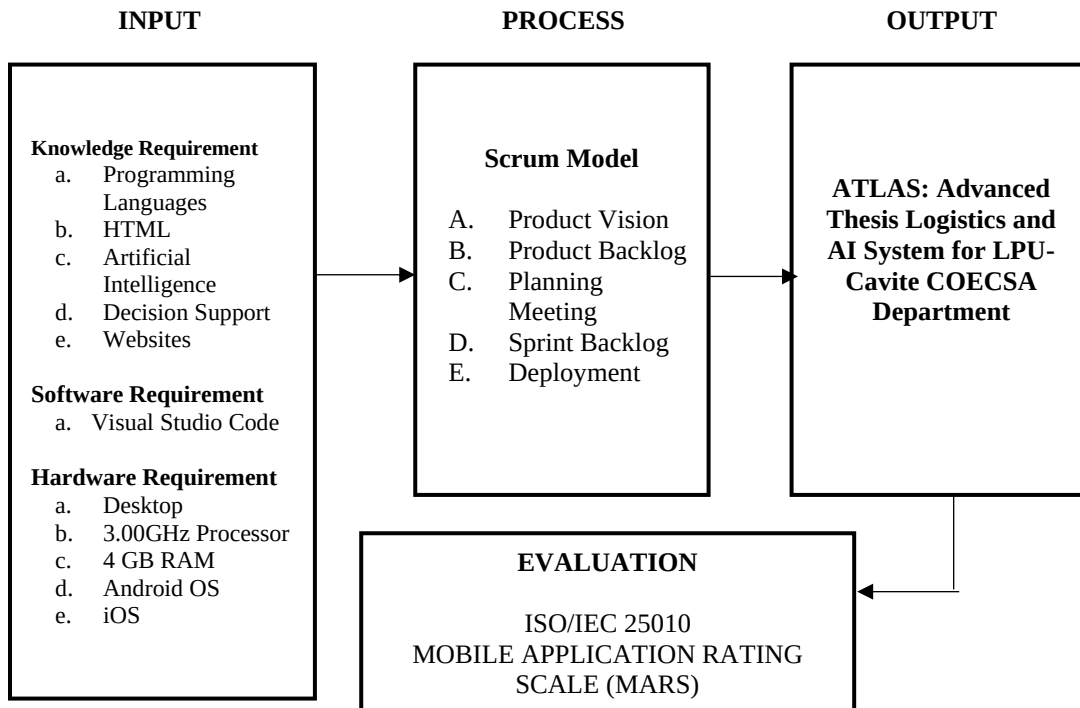
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	streamline the process, reduce conflicts, and provide intelligent assistance for both faculty and students CMS will serve as a crucial tool for managing academic content within ATLAS. It provides a flexible platform for faculty members to easily update rubrics, research titles, team lists, and panelist assignments without needing technical expertise. The CMS ensures that all important documents and data related to thesis projects are well-organized and easily accessible. By centralizing content management, the CMS will help streamline academic processes, reduce manual errors, and enhance the overall efficiency of handling research presentations and evaluations. Related Studies that Use Genetic Algorithm on Scheduling Genetic Algorithms (GAs) have emerged as a powerful tool for addressing various scheduling problems across different domains, particularly in production and job shop scheduling. Recent studies have demonstrated the versatility and effectiveness of GAs in optimizing scheduling systems by addressing constraints and enhancing performance metrics. One notable application of GAs is in the optimization of course scheduling systems. Chen's work highlights how GAs can be utilized to meet various constraints and optimization objectives in pharmaceutical laboratory course scheduling, emphasizing the importance of defining a fitness function to evaluate scheduling plans effectively (Chen, 2024). This approach not only improves scheduling efficiency but also adapts to the dynamic nature of course offerings and student needs.	
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	In summary, the application of Genetic Algorithms in scheduling systems is extensive and varied, with recent studies showcasing their effectiveness in optimizing course schedules. These algorithms not only improve scheduling efficiency but also adapt to complex constraints and dynamic conditions, making them invaluable in contemporary scheduling challenges.	
	Conceptual Framework The proponents undertaking the project must acquire knowledge in HTML, CSS, SQL, and PHP, as these skills are essential for the development of Decision Support System. The framework shows the system's components and their	
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relationship, focusing on how an AI-driven website processes can optimize scheduling and assessment within the academic environment.

Figure 1

Conceptual Framework of ATLAS.



Shown in Figure 1, the Agile Model has been chosen for it is suitable for the study, making it the preferred model for its flexibility and easily allows changes without affecting the project's timeline. Hence, it has been chosen for this study.

Figure 1 illustrates the project's framework representing the Input, Process, and Output phases of the study. The Input section shows the required Knowledge requirements, Software requirement, and Hardware requirements which are necessary for the proponents to develop the system. Next is the Process section, which the proponents follow throughout the development process of the system.

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	Finally, the Output section, which shows the outcome of the study, an AI-Driven System. Once the system has finished its development, it will undergo an evaluation using the ISO/IEC 25010 and MARS standard to evaluate its effectiveness and quality. Definition of Terms For the readers to fully comprehend this study, the following are the notable terms that have been utilized in the study: AI-Driven System. A system that is powered by Artificial Intelligence (AI) that will help automate processes like scheduling, data analysis, and decision-making, aimed to improving the efficiency and the accuracy in managing academic activities such as research presentations Decision Support System. Also known as DSS. A computerized system used to aid decision-making processes. In this project, the DSS helps students choose research topics and provides suggestions for thesis work by analyzing current trends and existing data, offering structured, objective recommendations. Scheduling System. A module in the system that will help automates the scheduling of the research defenses, matching the students and the panellists availability to avoid conflicts and ensure the smooth scheduling. Content Management System. This will allow the users to manage and change important elements like rubrics, team list, title list, and panelists data	
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providing a more flexible and user-friendly system for managing the academic content.

Thesis Topic Decision Tool. A component of the system that will use AI and data-driven algorithms to suggest the research topics based on the trends in academic research, helping the students to choose the relevant and unique topics.

Requirement Checker Tool. A feature of the system that will help the students meet the necessary requirements before their research defense. Tracking the completion of mandatory steps and alerting them when the requirements are still incomplete.

Pass Recommendation System. Decision tool will help the panellist to provide feedback and scores from the research defense, providing them a recommendation on whether the student should pass or fail based on the predefined criteria.

Research Title Acceptance Tool. It will help to analyze and compare the proposed research titles with the existing studies to ensure the uniqueness of the titles, avoiding redundancy, and reducing the risk of plagiarism.

Large Language Model. A type of AI model trained on large datasets to understand and generate human-like text. In this system, LLMs can be used for tasks like analyzing research titles, providing feedback, and generating suggestions based on trends and previous research.

Genetic Algorithm. is a search and optimization technique based on natural selection and evolution. It uses selection, crossover, and mutation to generate new

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	populations and converges towards optimal solutions. In this system, it can solve complex optimization problems like scheduling defenses and selecting research topics. It enhances efficiency, data-driven decisions, and automates time-consuming processes. ADD SA DULO KUNG SAN SYA GINAMIT DITO SA THESIS CHAPTER III METHODOLOGY	
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	Research Design In this study, the proponents used a quantitative survey research approach to evaluate and see the potential effectiveness of the ATLAS: Advanced Thesis Logistics and AI-System for COECSA of Lyceum of the Philippines University Cavite. This approach was chosen to get valuable data from students, advisers, and program chairs. The data will be analyzed to see and understand the relevance of the proposed system. To optimize scheduling, the Genetic Algorithm was chosen as the main algorithm for ATLAS. Genetic Algorithm is a robust optimization technique inspired by the process of natural selection. It is used in ATLAS to efficiently allocate limited slots for research presentations and to avoid conflicts, ensuring that all users of ATLAS have an optimized and organized schedule. Additionally, Gemini, a Large Language Model (LLM), is utilized within ATLAS to perform tasks related to natural language processing (NLP). Gemini aids in the text analysis of ATLAS, especially in sentiment analysis of feedback from the panelists, plagiarism detection in research titles, and suggesting thesis topics with emerging trends in computing. Sampling Technique Genetic algorithms (GA) are particularly suited to scheduling due to their adaptability and capability to explore large solution spaces with complex constraints, outperforming traditional heuristic algorithms like simulated annealing, expert systems, and ant colony algorithms. These alternative methods,	
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	although effective, often face limitations in managing the multi-dimensional and dynamic constraints of scheduling problems. For instance, simulated annealing relies on a cooling schedule that can be challenging to fine-tune for complex scheduling tasks, while expert systems require detailed rules that may not adapt well to dynamic changes in scheduling constraints. The ant colony algorithm, although effective in finding paths in discrete optimization problems, tends to require significant computational resources to handle large scheduling spaces and may converge more slowly. By contrast, the GA approach iteratively evolves solutions using adaptive operators selection, crossover, and mutation to enhance solution diversity and avoid local optima, making it highly effective for global optimization in scheduling. This study demonstrates that the improved genetic algorithm, enhanced with a local search operator and adaptive crossover and mutation rates, achieves faster convergence and higher-quality solutions. The experiments show that, compared to other algorithms, the GA yields better average fitness values and computational efficiency, making it an ideal choice for addressing the multi-objective demands of course scheduling. (Xu., 2021) Gemini, with its dual encoder model and memory-augmented mechanism, is uniquely positioned to excel in thesis management compared to other AI models. Unlike traditional algorithms, which may lack the ability to handle complex, multi-turn dialogues, Gemini’s architecture is specifically designed to understand and engage in sequential, context-rich interactions. This enables it to navigate the nuanced and evolving steps in thesis tasks, such as suggesting relevant topics, validating research titles, and providing pass recommendations. The model’s	
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	memory-augmented mechanism allows it to retain context across multiple dialogue turns, making it ideal for tasks that require consistency and continuity, such as following up on a student’s thesis development over time. Additionally, Gemini’s fine-grained dialogue state tracking ensures accurate and relevant responses by keeping track of the evolving context, which is essential in academic environments where clarity and adherence to specific guidelines are critical. While models like ChatGPT and LLaMA offer strengths in open-ended conversation and task-specific support, Gemini’s tailored approach to multi-step dialogue and task tracking makes it a superior choice for the specialized requirements of thesis management. (Krify., 2024) We chose to use Gemini over GPT, despite GPT being the leading LLM, due to our team's budget constraints. Given our financial limitations, Gemini was a more practical option for our project. Stratified random sampling will be employed to gather responses from Professors, Research Advisers, Research Panelists, Department Chairs, and Students through an online survey. This approach involves dividing the population into distinct subgroups (strata) based on their roles within the thesis process. A random sample will then be drawn from each subgroup, ensuring a proportionate and representative selection of participants. By capturing input from all relevant groups, this method ensures comprehensive feedback on the college’s thesis system and the potential benefits of an AI-driven decision support tool, while minimizing bias and enhancing the validity of the research findings.	
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Participants of the Study

The participant of this study includes three key groups from LPU-C COECSA Department: students, thesis advisers/professors, and the program chair. All the mentioned participants are vital in giving valuable feedback for potential improvements of ATLAS.

- Students will provide insights on how the system supports their scheduling, performance tracking and research title management
- Thesis advisors/professors will assess the system's ability to optimize scheduling, monitoring the students' progress, and improve the evaluation of research presentations
- The Program Chair will offer its insights into the system's overall impact on the department.

Research Locale

This study will be conducted at Lyceum of the Philippines University – Cavite (LPU-C), specifically within the College of Engineering, Computer Studies, and Architecture (COECSA). LPU-C is in General Trias, Cavite.

COECSA Departments has been chosen as it houses more complex research presentations and academic assessments. The participants will come from COECSA, as they are directly involved in research processes that our system aims to improve.

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	Research Instrument To gather data for this research, an online survey questionnaire using Microsoft Forms. The survey will target the key stakeholders including Professors, Research Advisers, Research Panelists, Department Chairs, and Students. The objective is to gain valuable insights into the current thesis system used within the college and to explore the potential of implementing an AI-driven decision support tool. Data Gathering Procedure The proponents relied on online surveys with the selected respondents. The survey was sent through email and can be accessed by using their LPU account. The questions for the survey were made based on the research objectives. The survey has categories depending on the respondents if either they are students, thesis professors/advisers and program chair. The respondents' privacy and confidentiality were strictly maintained throughout the data gathering process.	
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Multiple Constraints Analysis

Constraint	Impact on Scheduling	Impact on Functionality	Impact on Budget	Overall Impact
Panelist Availability	High	Medium	Low	High
Data Availability (for AI)	Medium	High	Medium	High
Developer Skill Levels	Low	High	Low	Medium
Budget Limitations	Medium	Medium	High	High
Internet Infrastructure	Low	Medium	Low	Medium

The ATLAS: Advanced Thesis Logistics and AI System for LPU-Cavite faces several constraints, including panelist availability, data availability, developer skill levels, budget limitations, and internet infrastructure. Panelist availability is crucial for efficient scheduling and can cause delays and conflicts. Data availability is essential for AI-driven tools, and it affects scheduling and functionality. Developer skill levels are also important, as they require high technical expertise for advanced features. Budget limitations can lead to delays and scaling back of features, but the overall scope and timeline are tied to available financial resources. Internet infrastructure moderately impacts functionality, especially for cloud-based or real-time data processing. Addressing internet infrastructure is relatively low, making it a manageable constraint in terms of budget. Overall, managing these constraints is essential for the success of the ATLAS project.

System Development Process

The proponents utilized the Scrum model for the development of ATLAS: Advanced Thesis Logistics and AI System for LPU-C. The Scrum model has 5 components that the proponents utilized. This includes Product Vision, Product Backlog, Planning Meeting, Sprint Backlog, and Deployment.

Figure 2. Scrum Model

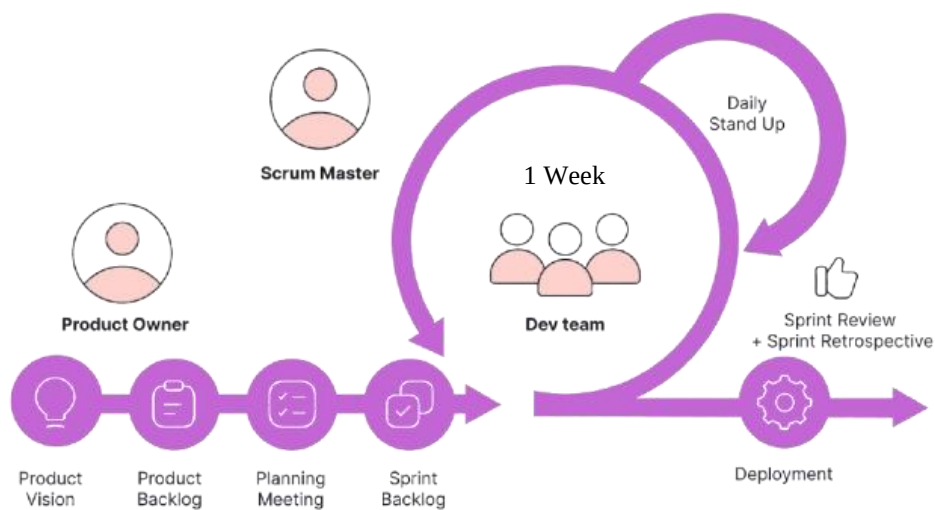


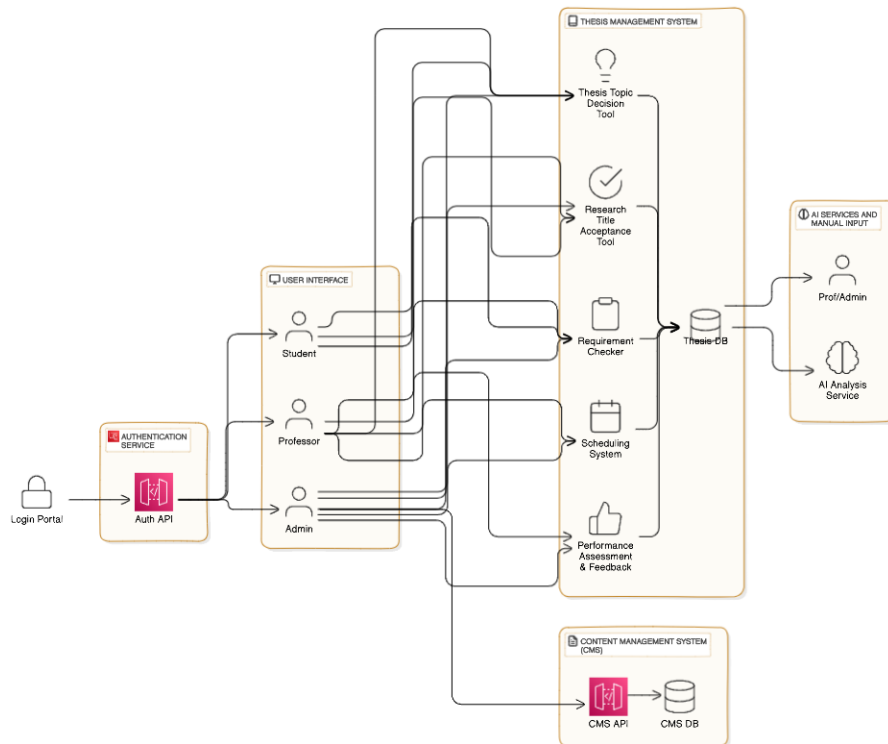
Figure 2 illustrates the process of Scrum Model. Below discusses each step pertaining to the actions of the life cycle:

Scrum Framework Overview

The Scrum model operates in short sprints typically lasting between 1-4 weeks. But for this study we intend to operate weekly. Each sprint consists of

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	planning and development. Key roles in Scrum includes Product Owner, Scrum Master, and Development Team, each roles plays a crucial role ensuring the success of the project. ▪ Product Owner: Responsible for defining the system features, managing backlogs, and prioritizing the systems' goals. ▪ Scrum Master: Facilitates the Scrum process, ensuring the team follows the cycle. ▪ Development Team: The team of developers, designers, and document writers that are responsible for building the project. System Architecture The users log in to access personalized dashboards based on their roles. Students input their interests and grades into the Thesis Topic Decision Tool, which leverages AI analysis to recommend relevant research topics. Upon proposing a title, the Research Title Acceptance Tool checks its uniqueness against an existing database. Students then the student will wait for the time of their defense, and the Scheduling System aligns schedules by querying availability to confirm a defense time. The Requirement Checker will be manual input by the prof/admin and the document submissions, alerting students of any pending requirements. Following the defense, Professor panelist provide feedback that Pass Recommendation System analyzes to classify student performance as "Pass" or "Fail." Lastly, admins utilize the Content Management System (CMS) to manage rubrics and panel assignments, ensuring all data remains updated and accessible.	
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Figure 3. Systems Architecture



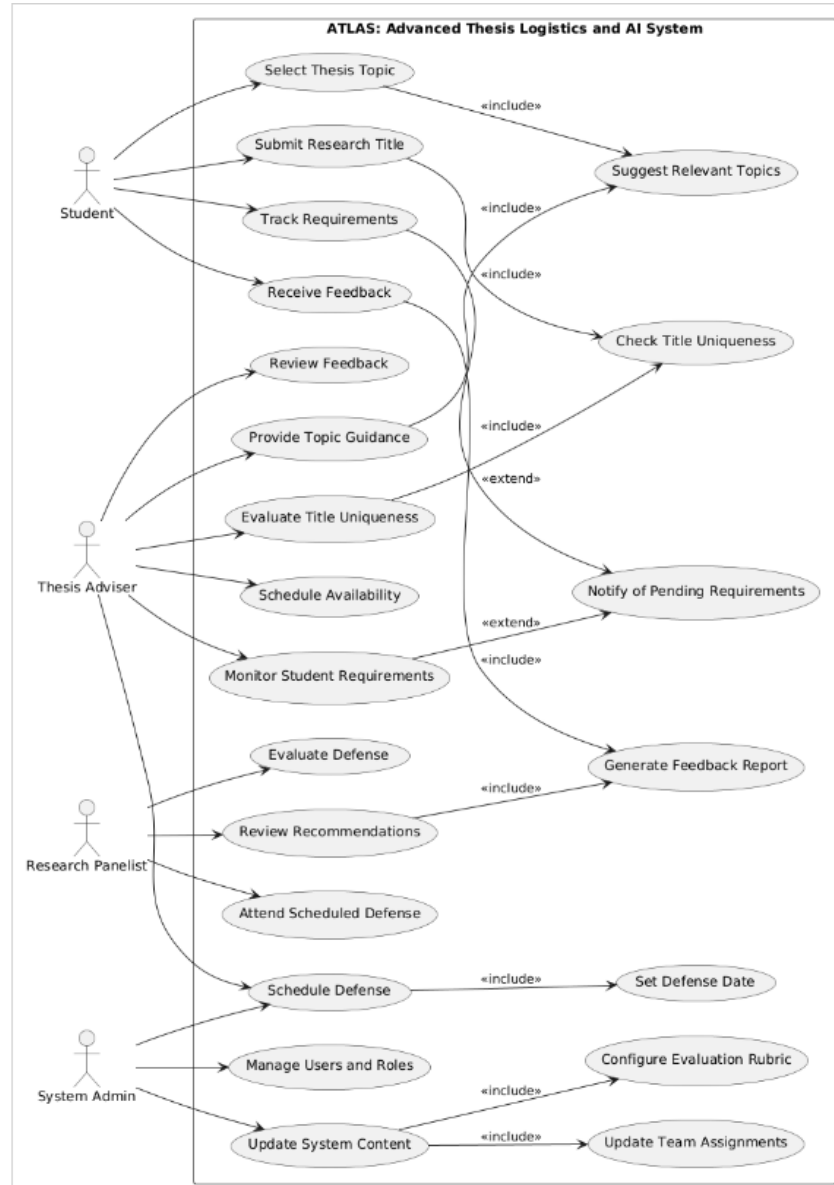
Object Modelling

Use Case

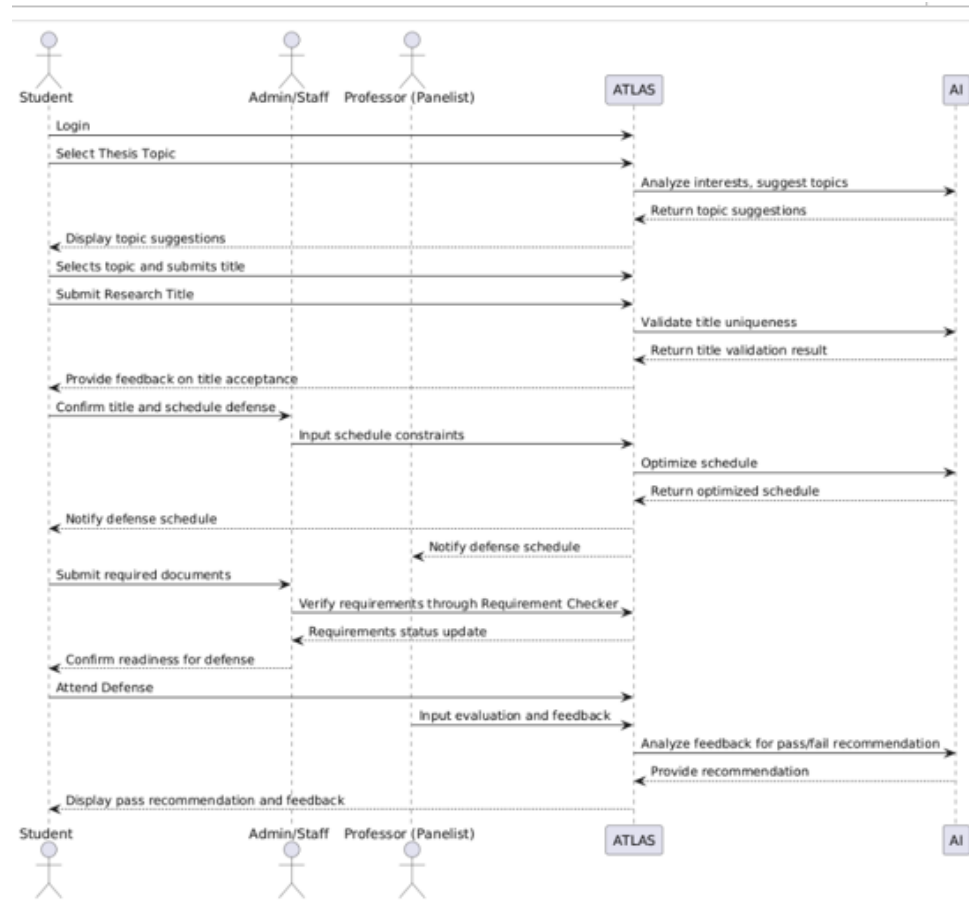
Students use tools for topic selection, title verification, and requirement tracking, while advisers guide students, schedule defenses, and monitor progress. Panelists attend scheduled defenses, provide evaluations, and review system-generated recommendations. Administrators manage roles, update system content, and maintain functionality. With AI-driven tools, ATLAS optimizes scheduling,

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feedback, and evaluation, reducing manual work and enhancing academic transparency and efficiency for all users.

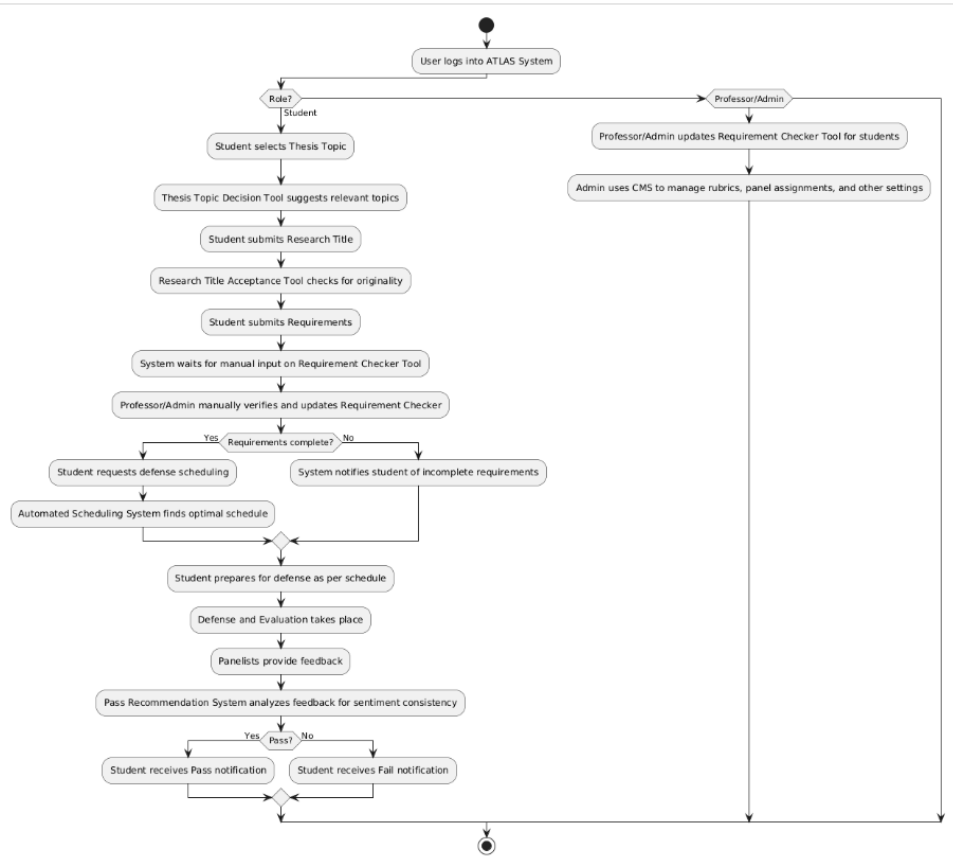


Sequence Diagram



The process begins with the student logging into ATLAS and using the AI-based thesis topic suggestion tool, which recommends relevant research topics based on their interests. Once a topic is chosen, the student submits a research title, which the AI system checks for uniqueness to ensure originality. After title approval, the admin coordinates the scheduling of the thesis defense, with AI optimizing the schedule based on participant availability, minimizing conflicts. In preparation for the defense, the admin uses the Requirement Checker to confirm

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	that all necessary documents are submitted. On defense day, the professor provides evaluation feedback, which the AI system analyzes to generate a preliminary pass or fail recommendation for the student. This automated, AI-supported approach not only ensures efficient scheduling and originality checks but also offers standardized feedback, enhancing transparency and efficiency throughout the thesis process. Activity Diagram Students and professors interact through a series of processes aimed at streamlining thesis management. When a student logs in, they can choose a thesis topic through the Thesis Topic Decision Tool, then submit their research title, which the system checks for originality. They then submit necessary requirements, tracked manually by professors and administrators through the Requirement Checker Tool. Once all requirements are met, students request a defense schedule, and the system’s scheduling algorithm assigns a conflict-free time slot. During the defense, panelists provide feedback, which the Pass Recommendation System analyzes for consistency, ultimately issuing a pass or fail result to the student. Professors and admins also use the system to update student requirement statuses and manage rubrics, panelist assignments, and content through the CMS, ensuring a smooth and transparent thesis process for all involved.	
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Data Analysis

20 students highlight key challenges in the research process. A significant portion 65% struggle with getting their research titles approved, and 70% have difficulty deciding on a research topic. This emphasizes the need for AI tools like the Research Title Acceptance Tool and Thesis Topic Decision Tool to streamline these areas. Additionally, 70% of students report problems submitting requirements on time, and 65% experience confusion or delays with defense schedules. AI-driven tools, such as a requirement tracker and an automated scheduling system, could

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address these issues effectively. For selecting research topics, 70% prefer AI-powered suggestion tools, and 80% use online resources like Google Scholar. To ensure title uniqueness, 55% favor AI tools, but 50% also value peer or mentor feedback, suggesting a hybrid approach would be beneficial. When managing document submissions, 45% prefer a manual checklist, while 40% each favor an AI-driven requirement tracker and faculty guidance. This indicates that combining AI with traditional methods may provide the most flexibility. In terms of evaluation tools, 55% find an AI-powered pass recommendation system very useful, while 75% consider a centralized system for managing defense requirements to be highly beneficial. Opinions on AI-generated performance evaluations are more mixed, with 45% finding them very useful but 25% preferring traditional methods, suggesting that AI should complement rather than replace human evaluation.

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Ethical Considerations

In the development of ATLAS: Advanced Thesis Logistics and AI-System for LPU-Cavite COECSA Department, the following considerations are needed to ensure the protection of all stakeholders and the responsible use of AI technology.

1. Data Privacy and Confidentiality

- The proponents ensures that all the data/information gathered throughout the duration of this project will be handled with strict confidentiality. The

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proponents ensures that unauthorized users will not have access to this data.

All participants will stay anonymous.

2. Informed Consent

- Before participating in this study, all participants are informed about the objectives and procedures involved. The proponents ensure that the participants fully understand, their rights and the voluntary nature of their involvement. Consent will be obtained through signed letters by the Dean of the department.

3. Intellectual Property and Plagiarism

- The system will respect and protect the intellectual property of both participants and the institution. Research titles, theses, and other academic content submitted by the institution and students will be safeguarded within the system. The plagiarism detector will ensure originality and have no similar existing titles.

5. Cybersecurity and Protection from Harm

- **Ensuring Security:** If the project involves software development, ethical consideration must include building secure and resilient systems to protect users from cyber threats.
- **Avoiding Harm:** Projects should avoid causing harm to individuals or systems, whether through negligence, vulnerabilities, or malicious design.\
- **Winston?**

6. Bias and Fairness

- To address potential algorithmic bias in ATLAS, specific attention was given to ensure fairness in the Genetic Algorithm used for scheduling and Gemini LLM used in Thesis Topic Decision Tool, Research Title Acceptance Tool, and Pass Recommendation Tool. Actions were made to prevent any biases that might favor groups or lead to unfair outcomes.
- Transparency is also a guiding principle in this study. A clear documentation was provided on how the proponents gathered data, used the data in this study, and analyzed the data within ATLAS.

7. Accountability and Integrity

- The researchers and developers of ATLAS will maintain accountability throughout the system's design, deployment and operation. The research team also ensures that ATLAS operates with the highest level of academic integrity, supporting fair evaluation practices.

8. Collaboration Ethics

- ATLAS encourages collaboration between students, faculty, and the administrative staff while upholding ethical standards in all interactions. Contributions made by all participants during the systems development will be acknowledged. The researchers and developers of ATLAS ensures that the system will be used as a tool for academic growth while promoting responsible conduct in research and academic evaluation.

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These ethical considerations ensure that an IT capstone project or research is conducted responsibly, safeguarding the interests of individuals, society, and the broader tech community.

CHAPTER IV

RESULTS AND DISCUSSION

**** Presentation of Project Design, Result of Testing, Evaluation and Validation, Analysis and Interpretation of Data**

Point of Discussion X (must be aligned to Objective X)

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Table 1

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Distance (m)	Pulse Width (Echo Time) (ms)
0.1	0.604
0.2	1.207
0.3	1.811
0.4	2.414
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Table 2

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Criteria	Mean	SD	Interpretation
quibusdam mollitia aut qua	3.93	0.25	Strongly agree
adipisci labore ut itaque	3.93	0.25	Strongly agree
excepturi ut sunt placeat et dolores	3.93	0.25	Strongly agree

Table 2. Continued...

Criteria	Mean	SD	Interpretation
praesentium et omnis temporibus	3.73	0.44	Strongly agree
adipisci labore ut itaque	4.00	0.00	Strongly agree
hic natus autem	3.20	0.40	Strongly agree
Grand Mean	3.79		

Note: For interpretation, the following remarks apply to mean interval: 4.00 – 3.26 for Strongly Agree, 3.25 – 2.51 for Agree, 2.50 – 1.76 for Disagree, 1.75 – 1.00 for Strongly Disagree.

The results presented in Table 2 illustrate the dolor sit amet. Aut nesciunt aliquam hic quos inventore aut aperiam deserunt non quia atque et deserunt dolorem. Non suscipit rerum non nesciunt aspernatur a iusto Quis. Hic velit magni est voluptatem sint et similique incidunt sed consequatur illo. Vel dolor dolorum ut voluptatum ipsum cum atque quisquam aut animi molestias. Est laborum inventore id nihil minima hic assumenda culpa aut animi voluptatem. Sit nesciunt accusamus id eius consequatur quo tempore neque ut dolorem magni.

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CHAPTER V

SUMMARY, CONCLUSION AND RECOMMENDATIONS

Summary and Conclusion

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Recommendations

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APPENDICES

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APPENDIX A

Certificate of Statistical Analysis

CERTIFICATE OF STATISTICAL ANALYSIS

Title of work: Characterization of Blast Resistance Gene in Rice (*Oryza sativa*)

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